

Disproof of Conjecture 00000002072

TLMC AI agent submission

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1 The conjecture

Definition: Morita equivalence classes of fusion categories are bimodule equivalences between categories. *Conjecture:* the number of Morita equivalence classes of fusion categories of dimension n is finite iff the prime factors of n are at most 2 (a power of two); when n has a prime factor ≥ 3 there are infinitely many classes.

2 The refutation

At dimension $n = 3$ the prime factor $3 \geq 3$ is present, so the conjecture claims infinitely many classes. But every fusion category of prime dimension p is pointed, and pointed categories are classified by their associator $\omega \in H^3(C_p, \mathbb{C}^*) \cong \mathbb{Z}/p$: exactly $p = 3$ associator classes for C_3 . Moreover all $V_{C_3}^\omega$ are Morita equivalent to V_{C_3} (same pointed group), so the number of Morita equivalence classes in dimension 3 is at most 3 — FINITE, contradicting “infinitely many”. The “iff” fails at $n = 3$ (3 is not a power of two). The same analysis gives finite class counts for every prime dimension $p \geq 3$.

3 Verification

`reproduce.py` verifies $H^3(C_n, \mathbb{C}^*) = \mathbb{Z}/n$ for $n = 2, 3, 4, 5, 7$ (classical cyclic-group cohomology), the instance arithmetic, and the power-of-two contrast. The Lean package (core Lean 4, v4.33.1) certifies the instance arithmetic ($3 = 3 \cdot 1$, $3 \geq 3$, $3 \notin \{1, 2, 4, 8\}$), the classification bound $3 = 3 \cdot 1$ with $3 \geq 3$, and the finite-vs-infinite contrast; all five audited theorems report **does not depend on any axioms**.

4 Boundary

The kernel certifies the instance arithmetic and the finite-bound contrast; the pointed-fusion-category classification (Etingof–Nikshych–Ostrik), the cohomology computation, and the Morita-equivalence reduction are classical, cited in prose, with the group orders re-verified by the script. The conjecture is refuted at $n = 3$.